

# Classification of Algebraic Varieties

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# Outline of the talk

## 1 Introduction

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- 2 The MMP for 3-folds

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  - The main Theorem
  - Curves
  - Surfaces
- 2 The MMP for 3-folds
- 3 Higher dimensional MMP

# Notation

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- If  $X$  is smooth of dimension  $n$ , then  $T_X$  denotes the **tangent bundle** of  $X$  and  $\omega_X = \Lambda^n T_X^\vee$  is the **canonical line bundle**.

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- $K_X$  denotes (a choice of) the **canonical divisor** i.e. a formal linear combination of codimension 1 subvarieties of  $X$  such that  $\omega_X = \mathcal{O}_X(K_X)$ .
- We would like to use  $\omega_X$  (or equivalently  $K_X$ ) to understand the geometry of  $X$ .

# Pluricanonical maps

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$$\phi_{\omega_X^{\otimes m}} : X \dashrightarrow \mathbb{P}^N$$

is defined by  $x \rightarrow [s_0(x) : \dots : s_N(x)]$ . (This map is not defined on the common zeroes of  $s_0, \dots, s_N$ .)

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- If  $X$  and  $X'$  are birational, then  $\forall m \geq 0$

$$H^0(X, \omega_X^{\otimes m}) \cong H^0(X', \omega_{X'}^{\otimes m}).$$

# Birational invariants

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- If  $\kappa(X) = \dim X$ , we say that  $X$  is of **general type**.
- One would like to understand the structure of this ring and to use its features to classify complex projective varieties.

# Finite generation

Today I would like to discuss the following result:

Theorem (Birkar-Cascini-Hacon-M<sup>c</sup>Kernan and Siu)

*Let  $X$  be a smooth projective variety, then  $R(X)$  is finitely generated.*

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The two proofs are independent. Our proof is algebraic and uses the ideas of the MMP (minimal model program). Siu's proof is analytic and requires  $X$  to be of general type.

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- $g = 1$ : There is a 1-parameter family of elliptic curves;  
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These form a  $3g - 3$ -dimensional family.  
 $K_X$  is ample and  $\phi_{\omega_X^{\otimes 3}}$  determines an embedding.
- $\omega_X^{\otimes 3} = \mathcal{O}_{\mathbb{P}^N}(1)|_X$  so that  $R(X)$  is finitely generated.

# Birational geometry of surfaces

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- The first problem in the birational classification of surfaces is that given any point on a surface  $x \in X$ , one can produce a birational morphism

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- The morphism  $\nu$  may be viewed as a surgery that replaces the point  $x$  by a rational curve  $E \cong \mathbb{P}^1 = \mathbb{P}(T_x X)$ .
- The exceptional curve  $E$  is called a  $-1$  **curve** since

$$\omega_{\tilde{X}} \cdot E = \deg(\omega_{\tilde{X}}|_E) = -1.$$

# Birational geometry of surfaces II

- By a Theorem of Castelnuovo, one may reverse this procedure i.e. given any surface  $X$  and any  $-1$  curve  $E \subset X$ , there exists a morphism  $\nu : X \rightarrow X_1$  such that  $X = \text{Bl}_x(X_1)$  for some point  $x \in X_1$ .

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- Note that  $\rho(X_1) = \rho(X) - 1 \geq 0$  and so this procedure can be repeated at most finitely many times.
- A **minimal surface** is a surface that contains no  $-1$  curves.
- By Castelnuovo's Theorem, given any surface  $X$  there exists a birational morphism to a minimal surface  $X \rightarrow X_{\min}$  (given by finitely many point blow downs).

# Birational geometry of surfaces III

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- If  $\kappa(X) \geq 0$ , then the minimal model  $X \rightarrow X_{\min}$  is unique and  $K_{X_{\min}}$  is nef. This means that for any curve  $C \subset X_{\min}$  we have

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- In fact one can show that  $\omega_{X_{min}}$  is **semiample** i.e. there is an integer  $m > 0$  such that the sections of  $H^0(\omega_{X_{min}}^{\otimes m})$  have no common zeroes.

# Birational geometry of surfaces IV

- Therefore,  $\phi_{\omega_{X_{min}}^{\otimes m}} : X_{min} \rightarrow \mathbb{P}^N$  is a morphism,  
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  - Threefolds
  - Running the minimal model program
- 3 Higher dimensional MMP

# MMP for 3-folds

- By work of Mori, Kawamata, Kollár, Reid, Shokurov and others it is possible to generalize this picture to 3-folds:

## Theorem

*Assume  $\dim(X) = 3$ . If  $\kappa(X) < 0$ , then  $X$  is covered by rational curves and  $R(X) \cong \mathbb{C}$ .*

*If  $\kappa(X) \geq 0$ , then there is a birational map  $X \dashrightarrow X_{\min}$  such that  $\omega_{X_{\min}}$  is nef (i.e.  $\omega_{X_{\min}} \cdot C \geq 0$  for any curve  $C \subset X_{\min}$ ).*

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- More precisely, if  $\kappa(X) \geq 0$ , there is a sequence of flips and divisorial contractions

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- More precisely, if  $\kappa(X) \geq 0$ , there is a sequence of flips and divisorial contractions

$$X \dashrightarrow X_1 \dashrightarrow \dots \dashrightarrow X_M = X_{\min}.$$

- It is known that  $\omega_{X_{\min}}$  is semiample (i.e.  $\phi_{\omega_{X_{\min}}^{\otimes m}} : X \rightarrow \mathbb{P}_{\mathbb{C}}^N$  is a morphism) and so  $R(X)$  is finitely generated.

# MMP for 3-folds II

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- Divisorial contractions are the analog of contracting a  $-1$  curve on a surface.
- Flips are a new operation that we will discuss below.
- The minimal model  $X_{min}$  is not unique.
- The varieties  $X_i$  have mild singularities.

# Running a MMP

- A **Minimal Model program** is a sequence of geometric operations  $X_i \dashrightarrow X_{i+1}$  that starting from a projective variety  $X$  produces a minimal model (if  $\kappa(X) \geq 0$ ) or a covering family of rational curves (if  $\kappa(X) < 0$ ).

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- We start with a projective variety  $X$  with mild singularities.
- If  $\omega_X$  is nef we are done ( $X = X_{min}$ ).
- If  $\omega_X$  is not nef, then consider the **cone of effective curves**

$$\overline{NE}(X) = \left\{ \sum r_i C_i \mid r_i \in \mathbb{R}^{\geq 0} \right\} / \equiv$$

(here  $C \equiv C'$  or  $[C] = [C']$  if for any line bundle  $L$  on  $X$ , we have  $L \cdot (C - C') = 0$ ).

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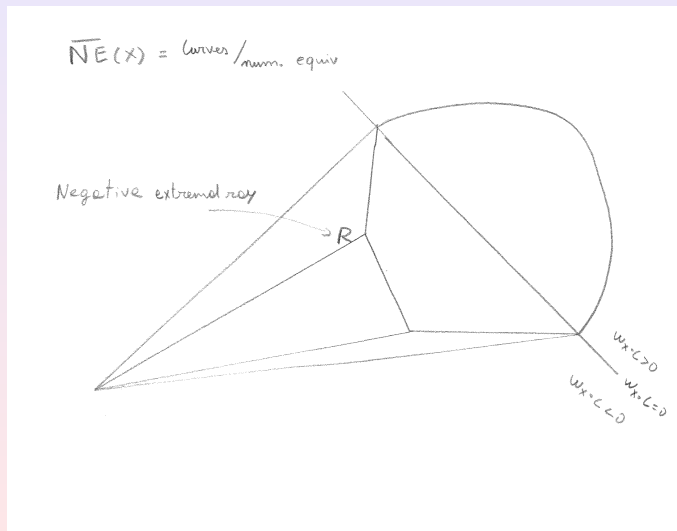
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- By the Cone Theorem, there is a negative extremal ray  $R = \mathbb{R}^+[C]$  where  $C \subset X$  is a curve such that  $\omega_X \cdot C < 0$  and there is a morphism  $\text{Cont}_R : X \rightarrow Z$  such that a curve  $D \subset X$  is contracted if and only if  $[D] \in R^+[C]$ .

# The Cone Theorem



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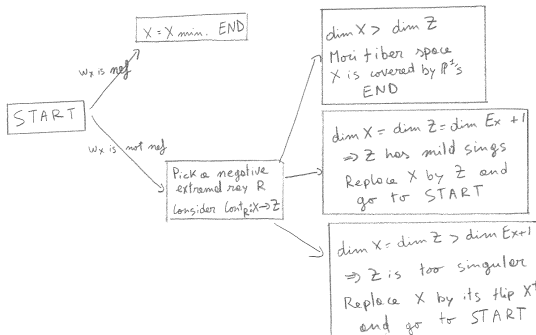
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- If  $\dim Z = \dim X$  and  $\text{Cont}_R$  is **small** (i.e.  $\dim \text{Exc}(X \rightarrow Z) \leq \dim X - 2$ ) then  $Z$  has “bad” singularities. In particular  $\omega_Z \cdot C$  does not always make sense.

# Running a MMP II

- If  $\dim Z < \dim X$ , then  $X$  is covered by  $\omega_X$  negative rational curves. (This only happens if  $\kappa(X) < 0$ .)
- If  $\dim Z = \dim X$  and  $\text{Cont}_R$  is **divisorial** (i.e.  $\dim \text{Exc}(X \rightarrow Z) = \dim X - 1$ ) then  $Z$  has mild singularities and we may replace  $X$  by  $Z$  (and restart the whole process).
- If  $\dim Z = \dim X$  and  $\text{Cont}_R$  is **small** (i.e.  $\dim \text{Exc}(X \rightarrow Z) \leq \dim X - 2$ ) then  $Z$  has “bad” singularities. In particular  $\omega_Z \cdot C$  does not always make sense.
- Instead of replacing  $X$  by  $Z$ , we will replace  $X$  by its flip  $X^+$ .

## MMP flow chart



# Flips

- The **flip** of  $X \rightarrow Z$  is a small birational morphism  $X^+ \rightarrow Z$  such that
  - 1)  $X \dashrightarrow X^+$  is an isomorphism on the complement of  $\text{Exc}(X \rightarrow Z)$ ,
  - 2)  $X^+$  has mild singularities, and
  - 3) If  $C \subset \text{Exc}(X^+ \rightarrow Z)$ , then  $\omega_{X^+} \cdot C > 0$ .

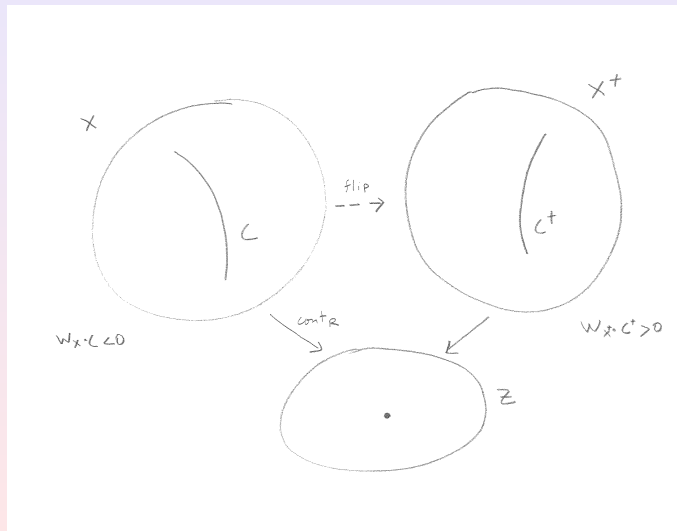
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- In other words, the flip  $X \dashrightarrow X^+$  is a surgery that replaces  $\omega_X$  negative curves by  $\omega_{X^+}$  positive curves.
- If the flip of  $X \rightarrow Z$  exists, it is unique and given by  $\text{Proj}_Z R(X)$ . So that to construct the flip we must show that  $R(X)$  is finitely generated over  $Z$  (i.e. if  $Z = \text{Spec} A$ ,  $R(X)$  is a finitely generated  $A$ -algebra).

# Picture of a flip



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- If  $X \rightarrow Z$  is a divisorial contraction, then  $\rho(Z) = \rho(X) - 1 \geq 0$ . If  $X \dashrightarrow X^+$  is a flip, then  $\rho(X^+) = \rho(X)$ . Therefore, it suffices to show that there is no infinite sequence of flips.

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- In dimension 3, this is relatively easy to do.

# Outline of the talk

- 1 Introduction
- 2 The MMP for 3-folds
- 3 Higher dimensional MMP
  - Reduction to log pairs of general type
  - Idea of the proof

# Log pairs

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- This is achieved by the adjunction formula which has the form  $(K_X + S)|_S = K_S + \Delta_S$  for some  $\mathbb{Q}$ -divisor  $\Delta_S$ .

# A Theorem of Fujino-Mori

Mori and Fujino have shown that:

## Theorem

*Let  $(X, \Delta)$  be any KLT  $\mathbb{Q}$ -factorial pair with  $\Delta \in \text{Div}_{\mathbb{Q}}(X)$ . Then there exists a KLT  $\mathbb{Q}$ -factorial pair  $(Y, \Gamma)$  of general type with  $\Gamma \in \text{Div}_{\mathbb{Q}}(Y)$  such that*

$$R(K_X + \Delta)^{(m)} \cong R(K_Y + \Gamma)^{(m)}$$

*for any sufficiently divisible integer  $m > 0$ .*

Recall that  $K_Y + \Gamma$  is of general type if  $\text{tr.deg.}_{\mathbb{C}} R(K_Y + \Gamma)^{(m)} - 1 = \dim Y$ .

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- If  $(X, \Delta)$  is KLT and  $K_X + \Delta$  is nef and of general type, then by the Base Point Free Theorem of Kawamata and Shokurov,  $K_X + \Delta$  is semiample.

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- As we have mentioned before, this implies that  $R(K_X + \Delta)$  is finitely generated.
- Therefore, to show that  $R(X)$  is finitely generated (for any smooth complex projective variety), it suffices to show that minimal models exist for KLT pairs of general type.

# Existence of minimal models for KLT pairs of general type.

## Theorem (Birkar-Cascini-Hacon-M<sup>c</sup>Kernan)

*Let  $(X, \Delta)$  be any KLT pair of general type. Then there is a minimal model  $f : X \dashrightarrow X_{\min}$  (i.e.  $K_{X_{\min}} + f_*\Delta$  is semiample).*

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*Flips exist (for KLT pairs; in any dimension).*

Remark: We are unable to show termination of flips. But we can pick a sequence of flips that terminates.

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- (Birkar-Cascini-Hacon-M<sup>c</sup>Kernan) If flips exist in dimension  $n + 1$  (and minimal models exist in dimension  $n$ ) then minimal models exist in dimension  $n + 1$ . (Uses the ideas of the MMP and in particular techniques developed by Shokurov.)

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- By adjunction, we may write  $(K_X + \Delta)|_S = K_S + \Delta_S$  where  $(S, \Delta_S)$  is KLT.
- Shokurov's observation is that to show that  $R(K_X + \Delta)$  is finitely generated over  $Z$  (and hence that the flip exists), it suffices to show that the restricted algebra

$$R_S(K_X + \Delta) := \text{Im}(R(K_X + \Delta) \rightarrow R(K_S + \Delta_S))$$

is finitely generated (over  $Z$ ).

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- The divisor  $\Theta$  is the limit of  $\mathbb{Q}$ -divisors  $\Theta_m$  defined by subtracting common components of  $\Delta_S$  and  $\text{Bs}(m(K_X + \Delta))/m$  from  $\Delta_S$ .

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- Using ideas of Siu, Kawamata and Tsuji we show that  $H^0(m(K_S + \Theta_m)) = \text{Im}(H^0(m(K_X + \Delta)) \rightarrow H^0(m(K_S + \Delta_S)))$ .

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$$H^0(m(K_S + \Theta_m)) = \text{Im}(H^0(m(K_X + \Delta)) \rightarrow H^0(m(K_S + \Delta_S))).$$
- The divisor  $\Theta$  is a priori an  $\mathbb{R}$ -divisor.
- Using a Diophantine approximation argument, we show that  $\Theta$  is in fact a  $\mathbb{Q}$ -divisor and  $\Theta = \Theta_m$  for some  $m > 0$ .

# The MMP with scaling

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- If  $t_1 = 0$  we are done. Otherwise we flip (resp. perform a divisorial contraction) along a ray  $R = \mathbb{R}^+[C]$  where  $(K_X + \Delta + t_1 A) \cdot C = 0$ .
- In this way, we obtain a sequence of rational numbers  $1 \geq t_1 \geq t_2 \cdots \geq 0$  and of flips/div. contractions  $X_i \dashrightarrow X_{i+1}$  such that  $X_i$  is a minimal model for  $(X, \Delta + t_i A)$ .

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- Note that to prove the above result on the finiteness of minimal models, we must use a compactness argument and hence it is critical to work with  $\mathbb{R}$ -divisors.